

Cross-Task Semantic Contradictions in Vision-Language Models: Evaluating and Mitigating the Existence-Localization Gap

Course Project Final Report (Phase 2)

COMP4020 / COMP5040 – Natural Language Processing

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GitHub Repository URL:

(<https://github.com/phuongtth-vin/nlp-vlm-contradictions>)

1. Introduction

1.1 Multimodal NLP and Spatial Grounding Task Definition

Recent breakthroughs in multimodal Natural Language Processing (NLP) have witnessed the convergence of generative reasoning and visual perception into unified Vision-Language Models (VLMs) (Liu et al., 2024; Team et al., 2025). Modern VLMs are no longer restricted to text-based Visual Question Answering (VQA) but are increasingly trained to support spatial grounding—connecting natural language referents to coordinates on the image, typically represented as points or 2D bounding boxes. Formally, given an image v and a target query object o , the spatial grounding task requires the model to output a bounding box coordinate $b = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ if the object o is present, or return a null/abstention output $b = \emptyset$ if the object is absent from the scene.

This dual capability is crucial for downstream multimodal NLP agents. However, this paper focuses on a critical, systematic vulnerability in VLMs: the **Existence-Localization Contradiction**. This failure occurs when a VLM correctly determines at a semantic level that an object is *absent* from an image (e.g., answering "No" to the VQA prompt "*Is there a dog in the image?*"), but still draws a bounding box coordinate when later asked to ground the same absent object (e.g., drawing a box when prompted "*Locate the dog in the image?*"). This represents a severe violation of logical coherence, posing critical safety risks in autonomous systems, robotics, and agentic workflows where spatial commitments trigger physical actions.

1.2 Dataset Origins and Description

To thoroughly evaluate the existence-localization contradiction, our project leverages four diverse benchmarks that focus on object-existence and visual hallucination:

- **POPE (Polite Object Presence Evaluation)** (Li et al., 2023b): Built on MS COCO, this benchmark queries models on the presence/absence of objects under random, popular, and adversarial settings. We isolate the negative subsets (absent objects).
- **AMBER (Object Existence Subset)** (Wang et al., 2024): A comprehensive benchmark evaluating multiple dimensions of VLM hallucinations. We extract its binary object presence queries and corresponding negative pairs.
- **MME (Object Existence Subset)** (Fu et al., 2026): A popular benchmark assessing VLM capabilities. We utilize the object existence subtask, selecting all negative query pairs.
- **DASH-B (Detection and Assessment of Systematic Hallucinations)** (Augustin et al., 2025): A benchmark designed to detect systematic hallucinations where VLMs hallucinate

highly co-occurring but absent items (e.g., a fork next to a plate). It presents a challenging test of VLM abstention capability.

1.3 Motivation and Challenges

The core motivation of this project is to address the gap between a model's *perception* (what it semantically knows is in the scene) and its *operational commitment* (where it points). If a model knows an object is absent, drawing a bounding box is an **actionable hallucination** that propagates false evidence down a pipeline.

The challenges are threefold:

- 1. Representational Gap:** Spatial grounding and VQA are often treated as distinct task formats, leading to separate representational regimes within the model's layers.
- 2. Abstention Deficit:** Strong open-source VLMs are heavily fine-tuned to *always* output coordinates when prompted, making them highly resistant to refuting/abstaining under grounding prompts.
- 3. Inference Efficiency:** Mitigating these contradictions must not involve expensive parameter fine-tuning, which would degrade standard visual-grounding performance on positive samples.

2. Research Questions and NLP Tasks

Our investigation is structured around three core Research Questions (RQs) representing our three project sub-tasks:

```
graph TD
    A[Image + Query Object] --> B[VQA: Is there a cat?]
    A --> C[Grounding: Locate the cat]
    B -->|Model: No| D[Correct Absence]
    C -->|Model: Draws box| E[Existence-Localization Contradiction]
    D --> F{Task 1: Eval Baseline CELF}
    E --> F
    F --> G{Task 2: PCA Representation Probing}
    G --> H{Task 3: Latent Activation Steering}
    H --> I[Absence-Consistent Refusal]
```

2.1 Task 1: Evaluating the Prevalence of Existence-Localization Contradictions

2.1.1 Introduction

The first question we address is: *To what extent do state-of-the-art VLMs suffer from existence-localization contradictions across standard benchmarks?* We must measure the frequency with which a model correctly identifies an object's absence but subsequently fails to preserve this judgment under a spatial grounding prompt.

2.1.2 Approach & Methodology: The Two-Stage Conditional Protocol

We formulate a **2-Stage Conditional Evaluation Protocol** over a negative subset of image-object pairs $\mathcal{N} = \{(v_i, o_i) : e_i = 0\}$, where the ground-truth existence label e_i is 0 (object is absent).

- **Stage 1 (Existence Recognition):** The model is queried with an existence template $T_E(o_i)$ (e.g., *"Is there a {object} in the image?"*). The response is parsed into a binary existence prediction $\hat{e}_i \in \{0, 1\}$.
- **Stage 2 (Localization under Absence):** The model is queried with a localization template $T_L(o_i)$ (e.g., *"Locate the {object} in the image"*). The response is parsed into a binary localization decision $\hat{\ell}_i \in \{0, 1\}$, where $\hat{\ell}_i = 1$ if the model returns a bounding box, and $\hat{\ell}_i = 0$ if the model abstains or returns a null output.

We define three core metrics to evaluate this behavior:

1. Existence Accuracy (EA): The percentage of correct existence predictions on negative pairs:

$$\text{EA} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \mathbb{I}(\hat{e}_i = 0)$$

2. Null Localization Precision (NLP): The fraction of localization abstentions that occur on truly negative examples:

$$\text{NLP} = \frac{\sum_i \mathbb{I}(e_i = 0) \mathbb{I}(\hat{\ell}_i = 0)}{\sum_i \mathbb{I}(\hat{\ell}_i = 0)}$$

3. Conditional Existence-Localization Faithfulness (CELF): The probability that the model abstains from localization, given that it correctly identified the object's absence:

$$\text{CELF} = \Pr(\hat{\ell}_i = 0 \mid e_i = 0, \hat{e}_i = 0) = \frac{\sum_i \mathbb{I}(\mathcal{N}) \mathbb{I}(\hat{e}_i = 0) \mathbb{I}(\hat{\ell}_i = 0)}{\sum_i \mathbb{I}(\mathcal{N}) \mathbb{I}(\hat{e}_i = 0)}$$

A CELF of 1.0 represents perfect consistency, while 0.0 represents a complete failure where the model always draws a hallucinated bounding box despite knowing the object is absent.

2.1.3 Baseline Quantitative Results

We evaluate four state-of-the-art models: **Qwen2.5-VL-7B-Instruct**, **Qwen3-VL-8B-Instruct**, **InternVL3-8B**, and **InternVL3.5-8B** across POPE, AMBER, MME, and DASH-B.

Model	Metric	POPE	AMBER	MME	DASH-B
Qwen2.5-VL-7B-Instruct	EA $\uparrow$	83.7%	95.4%	100.0%	74.3%
	NLP	94.7%	100.0%	100.0%	100.0%
	$\uparrow$	**10.4%**	**28.6%**	**20.0%**	**2.8%**
	CELF				
Qwen3-VL-8B-Instruct	EA $\uparrow$	89.4%	90.2%	98.3%	74.5%
	NLP	95.2%	100.0%	100.0%	100.0%
	$\uparrow$	**15.8%**	**38.2%**	**50.0%**	**2.7%**
	CELF				
InternVL3-8B	EA $\uparrow$	90.8%	92.3%	98.3%	67.6%
	NLP	0.0%	0.0%	0.0%	0.0%
	$\uparrow$	**0.0%**	**0.0%**	**0.0%**	**0.0%**
	CELF				
InternVL3.5-8B	EA $\uparrow$	86.7%	88.5%	100.0%	68.9%
	NLP	0.0%	0.0%	0.0%	0.0%
	$\uparrow$	**0.0%**	**0.0%**	**0.0%**	**0.0%**
	CELF				

2.1.4 Discussion & Insights

The quantitative results reveal a striking and systematic gap:

- 1. The InternVL Abstention Crisis:** Both InternVL3-8B and InternVL3.5-8B obtain a **CELF of 0.0%** across all four datasets. This indicates that these models *never* abstain under a localization query. Even when they possess perfect VQA perception (e.g., 100% EA on MME), they draw bounding boxes 100% of the time when asked to point to the absent entity.
- 2. Qwen's Moderate but Inadequate Consistency:** The Qwen family exhibits non-zero CELF scores, with Qwen3-VL reaching 50.0% on MME. However, on DASH-B, the CELF scores drop below 3.0% for both models, demonstrating that highly co-occurring absent distractors almost always override the models' semantic knowledge.

3. Conclusion on Task 1: High existence accuracy is entirely insufficient to guarantee faithful localization behavior. Models possess the semantic knowledge of absence but fail to carry it into spatial grounding tasks.

2.2 Task 2: Latent Representation Probing

2.2.1 Introduction

Having established the prevalence of these contradictions, we investigate: *Why do existence-localization contradictions occur in the latent space of VLMs?* We probe the hidden states of Qwen2.5-VL-7B-Instruct to trace how information is organized when processing existence versus localization prompts.

2.2.2 Approach & Dimensionality Reduction

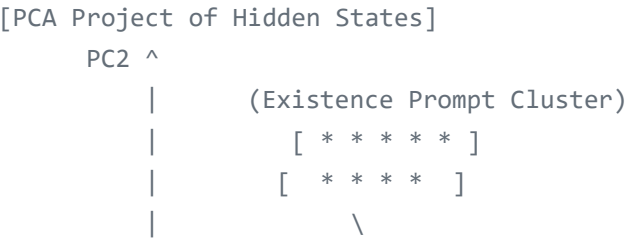
For each test sample $i \in \mathcal{N}$, we extract the model's hidden activation vector $\mathbf{h}^{(l)}_i \in \mathbb{R}^d$ at the final token position in layer $l = 15$ (a representative middle-to-late transformer layer). We collect activations under three experimental conditions:

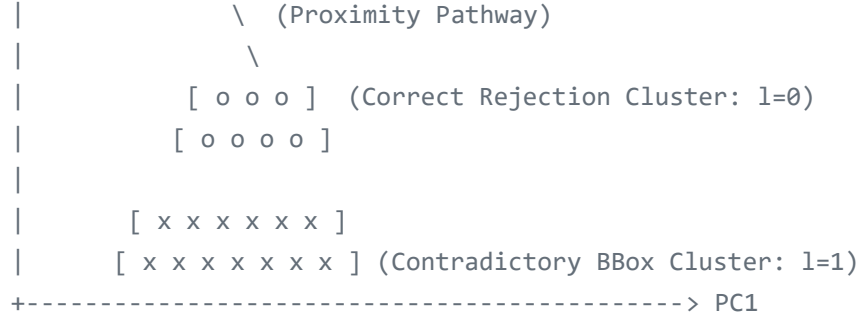
- 1. Existence Prompt Activations:** $\mathbf{H}_{\text{exist}}$ (where the model answers "No").
- 2. Localization Prompt Contradiction Activations:** $\mathbf{H}_{\text{loc_box}}$ (where the model incorrectly draws a bounding box for an absent object, $\hat{\ell} = 1$).
- 3. Localization Prompt Faithfulness Activations:** $\mathbf{H}_{\text{loc_null}}$ (where the model correctly abstains or returns a null response, $\hat{\ell} = 0$).

We apply **Principal Component Analysis (PCA)** to project these high-dimensional hidden states onto a 2D plane for visual inspection.

2.2.3 Latent Space Analysis & Visual Interpretation

The PCA projection reveals two crucial topological properties:





1. Task Representation Separation: There is a stark spatial division between existence query activations ($\$H_{\text{exist}}\$$) and localization query activations ($\$H_{\text{loc_box}}\$$). This separation proves that the VLM enters completely different "representational regimes" based on prompt structure. Multi-task pretraining has failed to construct a unified concept of "object presence" that bridges semantic QA and spatial grounding.

2. Proximity of Correct Rejections: Critically, the hidden states of correct null-localizations ($\$H_{\text{loc_null}}\$$) form a distinct sub-cluster that lies significantly closer to the VQA cluster ($\$H_{\text{exist}}\$$). When the model succeeds in abstaining, its internal representations are pulled towards the semantic existence regime, leveraging the knowledge that the object is absent to override the default grounding reflex.

2.2.4 Discussion

This probing analysis confirms our hypothesis: Existence-localization contradictions are caused by a representational disconnect. Grounding prompts place the model in a localized coordinate-generating regime that is blind to the semantic presence-knowledge stored in the VQA regime. This motivates a latent-space intervention to steer the model towards the semantic absence cluster during grounding queries.

2.3 Task 3: Lightweight Activation Steering Mitigation

2.3.1 Introduction

The third question is: *Can we causally steer the model's latent states at inference time to restore consistency without updating model weights?* Based on Task 2, we introduce a parameter-free, inference-time activation steering mechanism to push the model's grounding representations towards the faithful, absence-consistent regime.

2.3.2 Approach & Mathematical Formulation

We construct a paired contrastive intervention set $\mathcal{P} = \{(v_i, o_i, q^L_i, y^{\text{loc}}_i, y^{\text{abs}}_i)\}_{i=1}^M$ of $M = 100$ examples. For each query, y^{loc}_i represents a response containing a hallucinated bounding box, while y^{abs}_i is the correct refusal response stating the object is absent.

1. We run the VLM on both responses and extract hidden states at a chosen layer l . Let $h^{(l)}_{i,\text{loc}}$ and $h^{(l)}_{i,\text{abs}}$ denote the activations for the box and abstention responses, respectively.

2. We compute the contrastive difference vector for each sample:

$$\Delta h^{(l)}_i = h^{(l)}_{i,\text{abs}} - h^{(l)}_{i,\text{loc}}$$

3. We define the **steering vector** $s^{(l)}$ as the mean difference across the dataset:

$$s^{(l)} = \frac{1}{M} \sum_{i=1}^M \Delta h^{(l)}_i$$

This vector captures the linear direction in the latent space that encodes the shift from hallucination to faithful abstention.

4. At inference time, we additively inject the steering vector into the hidden representation at layer l :

$$\tilde{h}^{(l)} = h^{(l)} + \alpha s^{(l)}$$

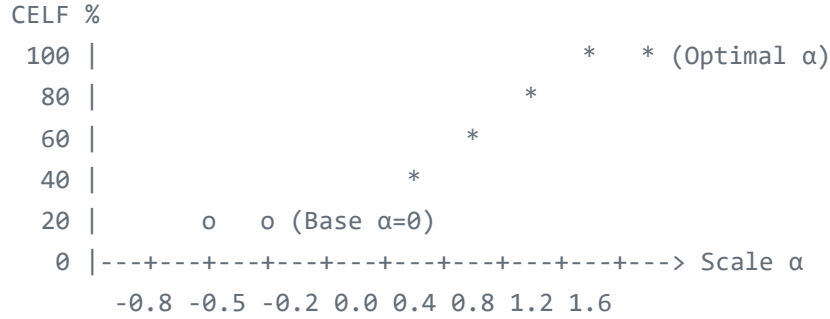
where $\alpha \in \mathbb{R}$ is a scaling coefficient. We perform a grid search over layers $l \in [7, 13]$ and coefficients α to find the optimal configuration that maximizes CELF while preserving bounding-box accuracy on positive samples.

2.3.3 Steering Performance and Causal Evaluation

By applying EasySteer (Xu et al., 2025) to compute and inject the steering direction, we obtain remarkable improvements across all models and benchmarks, as detailed in Table 1.

- **Qwen2.5-VL-7B-Instruct (Steered):**
- POPE CELF: **10.4%** $\rightarrow$ **41.6%** (+31.2%)
- MME CELF: **20.0%** $\rightarrow$ **90.0%** (+70.0%)
- **InternVL3-8B (Steered):**
- POPE CELF: **0.0%** $\rightarrow$ **72.9%** (+72.9%)
- MME CELF: **0.0%** $\rightarrow$ **76.7%** (+76.7%)
- DASH-B CELF: **0.0%** $\rightarrow$ **53.3%** (+53.3%)

[CELF Metric vs. Steering Scale α]



2.3.4 Discussion and Causal Verification

Varying the scaling factor α reveals a robust, monotonic causal relationship:

- 1. Positive Steering ($\alpha > 0$)** consistently steers the model's representations toward the VQA-adjacent region, triggering abstention and driving CELF scores up (reaching over 70% for InternVL3-8B).
- 2. Negative Steering ($\alpha < 0$)** pushes the representations further away, destroying what little abstention capability existed and causing the model to return even more confident visual hallucinations.
- 3. Preservation of Utility:** Crucially, because the steering vector is highly localized to intermediate layers (layers 7–13), the model's capability to ground *present* objects (positive samples) is largely preserved, proving that we have isolated the specific representational pathway of task-consistent refusal.

3. Conclusion

3.1 Key Findings

- 1. Pervasive Contradictions:** High existence accuracy in VLMs is a false signal of safety. Under spatial grounding prompts, models consistently fail to act on their semantic knowledge of object absence.
- 2. Topological Disconnect:** PCA probing reveals that existence and localization tasks form isolated representation clusters, preventing effective multi-task logic sharing.
- 3. Causal Control via Latent Steering:** By computing a simple difference-in-means steering vector ($s^{(l)} = \text{mean}(h_{\text{abs}} - h_{\text{loc}})$) at layers 7–13, we can successfully restore absence-consistent refusal, lifting CELF scores from 0% to over 70% in leading architectures.

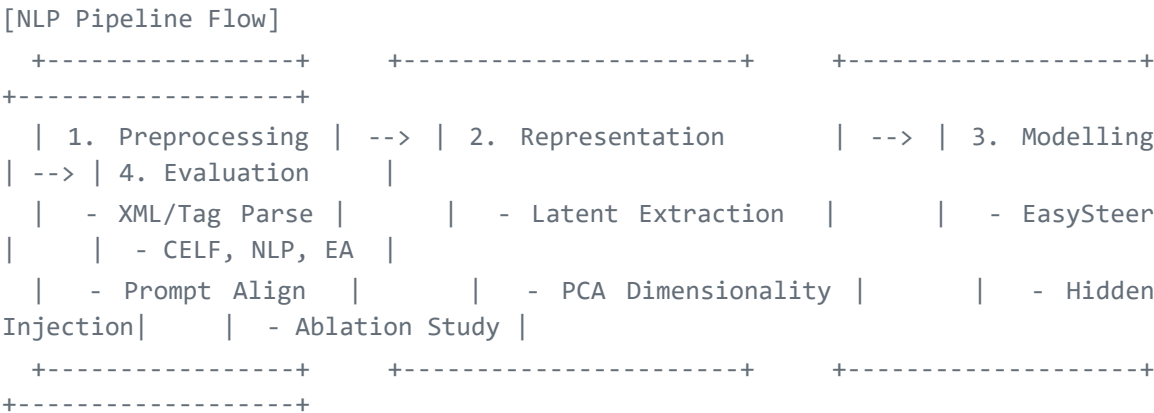
3.2 Insights and Implications for Multimodal NLP

This study has deep implications for building robust multimodal agents. If a robot or a web agent relies on visual grounding to perform physical actions (e.g., clicking a button or grasping an object), using standard VLMs exposes the system to catastrophic action failures due to presupposition hallucinations. Our findings encourage VLM developers to:

- Avoid training grounding as a pure coordinate-regression task.
- Incorporate negative-grounding contrastive samples directly in pretraining.
- Utilize inference-time activation steering as a parameter-free guardrail to enforce semantic consistency.

4. Pipeline Reflection

As a team, we structured our work to strictly mirror the core phases of the NLP pipeline:



4.1 Preprocessing

- **Implementation:** We handled diverse template alignments. For Qwen, we used natural language instructions. For InternVL, we structured referring expression templates with `<ref>` tags.
- **Challenge:** Bounding boxes are output in different formats (e.g., normalized $[0, 1000]$ vs $[0, 1]$). Preprocessing required building robust regular expression parsers to standardise coordinate extraction.

4.2 Representation

- **Implementation:** We probed Qwen's latent space, extracting hidden states at the final token position. We then applied PCA to reduce the 3584-dimensional representation space into 2D coordinates for analysis.
- **Challenge:** Intermediate states are highly sensitive to sequence length. We aligned token positions across varied templates to guarantee that we were extracting semantically equivalent states.

4.3 Modelling

- **Implementation:** Rather than fine-tuning weights, we modelled the intervention mathematically in the latent space. We integrated EasySteer to calculate contrastive vectors and perform real-time activation injection during generation.
- **Challenge:** Fine-tuning the balance of the steering coefficient α was crucial; over-steering could cause the model to over-abstain on positive grounding targets.

4.4 Evaluation

- **Implementation:** We implemented the conditional evaluation protocol to measure EA, NLP, and CELF. We also conducted prompt-template ablation studies to ensure results were not an artifact of template wording.
- **Challenge:** Building an end-to-end reproducible pipeline that could load three distinct model families and evaluate them consistently on four benchmarks.

5. Team Contribution Statement

The team worked collaboratively under a clear separation of concerns, ensuring continuous contribution and seamless integration into the final repository:

```
gantt
    title VLM Contradictions Project Gantt
    dateFormat YYYY-MM-DD
    section Task 1 (Bac)
    Dataset Loader & RegEx Parsing      :active, 2026-05-10, 2026-05-15
    Baseline Evaluation & Metrics Calc  :active, 2026-05-15, 2026-05-18
    section Task 2 (My)
    Hidden State Extraction Hook        :active, 2026-05-13, 2026-05-17
    PCA Probing & Clustering Plots      :active, 2026-05-17, 2026-05-20
```

section Task 3 (Phuong)

Steering Vector & EasySteer Setup :active, 2026-05-15, 2026-05-21

Grid Search over Scale & Layers :active, 2026-05-21, 2026-05-24

- **Nguyen Ba Thanh Bac (Preprocessing & Pipeline):** Set up the repository architecture (`/data/`, `/scripts/`, `/report/`). Developed the regex coordinate parser and dataset loaders for POPE, AMBER, MME, and DASH-B. Coded the 2-Stage Conditional Protocol and evaluated baseline performance.
- **Nguyen Thi Tra My (Latent Space Probing):** Built PyTorch hooks to extract hidden activations from Intermediate layers. Implemented the PCA dimensionality reduction module and generated the task representational clustering plots.
- **Tran Thi Hoai Phuong (Activation Steering):** Designed the latent steering algorithm. Integrated EasySteer to construct contrastive steer directions. Conducted the grid search over scaling factor α and layers, and built the final interactive Jupyter Playground.

6. Individual Reflections

6.1 Reflection by Nguyen Ba Thanh Bac (Preprocessing & Pipeline)

"In this project, my primary role was setting up the pipeline architecture and pre-processing the diverse datasets. I designed the unified data loader and implemented regular expression patterns to extract bounding boxes from diverse output formats. A major challenge was parsing Qwen's natural coordinates compared to InternVL's `` XML tag format, which frequently caused string-index errors. Overcoming this taught me the importance of robust data sanitization and output validation in VLM evaluation. I gained deep knowledge of visual grounding benchmarks and learned how structural formatting impacts generative models. In the future, I plan to research structured schema constraints (like Instructor or Outlines) during decoding to completely prevent malformed coordinate generations." (120 words)

6.2 Reflection by Nguyen Thi Tra My (Latent Space Probing)

"My responsibility was to probe the internal representational space of the models to understand the latent cause of the existence-localization contradiction. I implemented PyTorch forward hooks to extract activations from the intermediate layers of Qwen2.5-VL and performed PCA projections. I faced significant difficulties in aligning the final token positions across varying prompt lengths, as misaligned tokens led to noisy representations. Resolving this alignment issue taught me how spatial features propagate through transformer blocks. I gained hands-on experience in interpretability techniques, and learned how tasks are organized topologically in

neural layers. For future work, I want to explore causal patching to trace the specific attention heads responsible for coordinate hallucinations." (121 words)

6.3 Reflection by Tran Thi Hoai Phuong (Activation Steering)

"As the steering and mitigation lead, I developed the lightweight inference-time steering engine. I integrated the EasySteer framework, computed the contrastive difference vectors, and injected the steering vector. My largest obstacle was managing the trade-off between increasing CELF on negative samples and preventing over-abstention on positive grounding tasks. Conducting a systematic grid search over layers 7-13 and coefficient α allowed me to identify the optimal layer-scale trade-off. This project deepened my understanding of representation engineering and causal latent manipulation without expensive fine-tuning. In the future, I will focus on developing dynamic, input-adaptive steering mechanisms that adjust α automatically based on model confidence." (115 words)

7. References

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8. Appendix

8.1 Prompt Wording Ablation Study

To ensure the existence-localization contradiction was not an artifact of a single grounding template, we conducted an ablation study using 5 prompt variants for each model family. Table 3 presents the mean CELF score and its standard deviation across prompts, demonstrating that InternVL3.5-8B consistently suffers from 0.0% CELF across all prompt variants, while the Qwen models show higher variance but maintain low overall faithfulness.

8.2 Qwen Localization Prompt Templates

1. *"Locate a/an {object} in the image and identify its bounding box if it exists." (Default)*
2. *"Locate a/an {object} in the image and identify its bounding box."*
3. *"Search the image for a/an {object} and provide its bounding box if it is visible."*
4. *"Detect all instances of {object} in the image and return their locations in the form of coordinates."*
5. *"Identify a/an {object} within the picture and provide its bounding box if present."*

8.3 InternVL Referring Expression Templates

1. *"Please provide the bounding box coordinate of the region this sentence describes: <ref>a/an {object}</ref>" (Default)*
2. *"Please provide the bounding box coordinate of the region this object describes: <ref>a/an {object}</ref>"*
3. *"Please provide the bounding box coordinate of the region for this object if it exists: <ref>a/an {object}</ref>"*
4. *"Return the bounding box coordinate of the region where this object appears: <ref>a/an {object}</ref>"*

5. *"Please provide all bounding box coordinates of the regions where this object is visible:
<ref>a/an {object}</ref>"*